EcoSort

Waste Recognition and Sorting Assistant

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ITAI 1378 - Computer Vision and AI

Midterm Project Blueprint | Tier 1



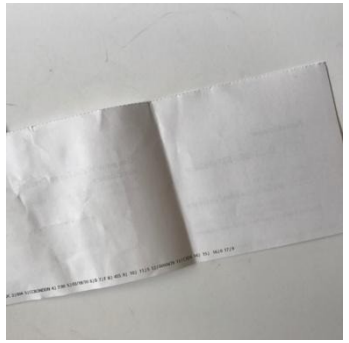
Cardboard



Glass



Metal



Paper



Plastic



Trash

Photos: TrashNet dataset (Thung & Yang, 2016)

The Problem

- Many people are not sure which bin an item goes in.
- Some things look recyclable, but they are not.
- Items in the wrong bin can contaminate recycling and make it work less well.
- Rules are different in different cities, so it gets more confusing.
- Who has this problem: households, students, and anyone sorting trash at home.
- EcoSort will give a quick first guess from a photo, plus a simple tip.



Paper or trash? (The dataset says trash.)



Clear plastic box: which bin?

The Solution

EcoSort is a small app. You upload a photo of one waste item, and it tells you the type of waste and how to sort it.



1. User uploads a photo

→ **2. Model looks at the photo**
MobileNetV2 (image classification)

→ **3. App shows the category**
Example: "Plastic" + confidence %

→ **4. App shows a sorting tip**
Example: "Empty it and check your local rules."

- The app works with single photos only, not live video.
- It sorts into 6 types: cardboard, glass, metal, paper, plastic, trash.
- It is for learning and basic help. It is not a robot or a smart bin.
- Take photos in a safe place, not near traffic.

Technical Approach

- Technique: image classification
  - Each photo shows one item, so the model only needs to give one label.
- Model: MobileNetV2 with transfer learning
  - It is already trained on ImageNet. I will add a new last layer with 6 outputs and train it on waste photos.
  - It is a small model (about 3.5 million parameters), so it should train fast.
- Language and framework: Python with TensorFlow/Keras
- Where I will work: Google Colab (free)
- App: a simple Streamlit page to upload a photo, if I have time
- Why this fits: it is simple, free, and doable in one term.



Input: a photo like this

↓



Resized to 224 × 224 pixels before it goes into the model

Data Plan

- Dataset: TrashNet
  - github.com/garythung/trashnet
- 2,527 photos, 6 labels (I counted the files myself)
- Each photo shows one item on a white background.
- Labels come from the folder names.
- Split: 70% train, 15% validation, 15% test
  - I will only use the test set at the end, so the result is fair.
- Limitation: the photos are very clean, and "trash" has only 137 photos.

Label	**Photos**
Cardboard	403
Glass	501
Metal	410
Paper	594
Plastic	482
Trash	137
Total	**2,527**

Source: TrashNet (Thung & Yang, 2016), MIT License

Link: https://github.com/garythung/trashnet


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Success Metrics

These are my goals. I have not tested anything yet.

- Main goal: at least 85% accuracy on the test set
  - How I will measure it: correct answers ÷ all test photos.
  - The test photos are never used for training.
  - I will also make a confusion matrix to see which types get mixed up.
- Second goal: less than 2 seconds per photo
  - How I will measure it: the average time from upload to answer.
  - I will write down what computer I used (Colab or the Streamlit app).
- For comparison: the TrashNet authors got about 75% with their own model. I hope transfer learning will do better.



Test photos will look like these. The model will not see them during training.

Milestone Plan

Phase	When	Goal	Milestone
1. Blueprint	Week 10	Plan the project	Slides, README, data README, AI log
2. First Working Demo	Week 11	Run the pretrained model on a few photos	The model gives an answer for each photo
3. Make It Yours	Weeks 12-13	Train the model on TrashNet	The model predicts the 6 waste types
4. Improve and Measure	Week 14	Make it better and test it	Accuracy, speed, and mistakes written down
5. Package and Present	Week 15	Build the app and finish the repo	Demo video, final slides, GitHub repo







Risks and Resources

- Risk 1: The model may do badly on real photos (blurry, strange angles, messy backgrounds).
  - Plan B: add data augmentation, keep only the 6 types, and show the mistakes honestly.
- Risk 2: Setup problems, slow training, or Colab limits may slow me down.
  - Plan B: use smaller images, a smaller part of the data, and a notebook demo if the app is late.





Hard photos: a bottle from the top and a can close up. Real photos can be even harder.

Resources

- Python
- TensorFlow/Keras
- Google Colab
- TrashNet dataset
- Streamlit (if I have time)

Cost: $0

All tools are free. Colab has some usage limits.